

Novel Contributions

- Formulated a leak-free, risk-aware temporal modeling framework addressing evaluation bias and interpretability gaps in deep financial forecasting.
- Advanced hybrid sequence architectures by decoupling temporal representation learning from regression decisions.
- Introduced local–global temporal fusion to jointly model short-horizon volatility structures and long-horizon dependencies.
- Reframed attention mechanisms as temporal attribution tools, enabling explainable and trustworthy sequence predictions.
- Designed reproducible experimental pipelines aligned with real-world deployment constraints.